

POLITECNICO
MILANO 1863

# Best Bike Paths – RASD Document

RASD document for the
Software Engineering 2 course assignment

Authors: **Alessandro Burato, Gabriele Crimella, Davide Faccini**

Teacher: Elisabetta Di Nitto
Academic Year: 2025-26

# Contents

**Contents** **i**

**1 Introduction** **1**
1.1 Purpose . . . 1
1.1.1 Goals . . . 1
1.2 Scope . . . 2
1.2.1 World Phenomena . . . 2
1.2.2 Machine Phenomena . . . 3
1.2.3 Shared Phenomena . . . 3
1.3 Definitions, Acronyms, Abbreviations . . . 4
1.4 Reference Documents . . . 4
1.5 Document Structure . . . 4

**2 Overall Description** **5**

**3 Specific Requirements** **7**
3.1 Functional Requirements . . . 7
3.1.1 Requirements . . . 7
3.1.2 Use Cases Diagrams . . . 9
3.1.3 Use Cases . . . 9

**4 Formal Analysis Using Alloy** **29**
4.1 Signatures . . . 29
4.1.1 Status . . . 29
4.1.2 Segment . . . 29
4.1.3 Path . . . 30
4.1.4 Statistics . . . 30
4.1.5 Trip . . . 30
4.1.6 RegisteredUser . . . 31
4.2 Facts . . . 31

4.2.1 Segment facts . . . . . 32
4.2.2 Path facts . . . . . 32

**5 Effort Spent** **33**

**6 References** **35**

**List of Figures** **37**

**List of Tables** **39**

# 1 Introduction

## 1.1. Purpose

The primary purpose of this project, initiated by a leading Bike Association, is to develop and deploy an advanced support system for the cycling community designed to thrive within a transport landscape dominated by cars.

The core idea is to foster a collaborative digital environment where cyclists can actively record, consult, and publish both their own and other users' riding paths.

This platform will serve as the definitive centralized resource for finding highly recommended routes tailored precisely to the user's specific experience, preferences, and safety considerations.

Ultimately, the system aims to significantly enhance the quality of the cycling experience by providing verified, community-rated routes, thereby connecting riders and making high-quality path discovery universally accessible

### 1.1.1. Goals

Here are the goals that BBP has to fulfill. BBP has to:

| Goal | Description |
|---|---|
| G1 | Recognize suitable bike paths |
| G2 | Allow the user to browse different bike paths |
| G3 | Allow the user to see bike paths statistics |
| G4 | Allow the user to search for paths between two locations |
| G5 | Allow the user to see paths optimality based on other registered users' reviews |
| G6 | Allow user to see weather information when available |
| G7 | Allow users to register to the application using credentials |
| G8 | Allow registered user to review published bike paths |
| G9 | Allow registered user to confirm or reject false positives in track recording |
| G10 | Allow registered user to let the system automatically record suitable bike paths |
| G11 | Allow registered user to manually create suitable bike paths |
| G12 | Allow registered user to publish bike paths |
| G13 | Allow the store of bike paths data for registered users |
| G14 | Allow registered user to add publishable information about bike paths |

Table 1.1: List of Goals

## 1.2. Scope

### 1.2.1. World Phenomena

| WP | Description |
|---|---|
| WP1 | A user physically moves through space using a bicycle |
| WP2 | The physical conditions of a bike path or street change (e.g., presence of potholes, road maintenance status, or obstacles) |
| WP3 | The meteorological conditions in the geographic area vary (e.g., temperature, wind speed) |
| WP4 | A user intends to reach a specific destination starting from a specific origin point |

Table 1.2: World Phenomena

### 1.2.2. Machine Phenomena

| MP | Description |
|---|---|
| MP1 | The system guesses that the user is biking by analyzing the detected speed |
| MP2 | The system computes statistics related to a trip (e.g., total distance covered, average speed, performance metrics) |
| MP3 | The system identifies potential anomalies (e.g., potholes) by processing raw data from the device sensors (accelerometer and gyroscope) |
| MP4 | The system computes the score of a path based on its status and its effectiveness in reaching the destination |
| MP5 | The system merges path information incoming from multiple users, evaluating data freshness and the number of confirmations received |
| MP6 | The system stores trip history and path information in the database for future retrieval |

Table 1.3: Machine Phenomena

### 1.2.3. Shared Phenomena

**World-Controlled Phenomena**

| SP_WC | Description |
|---|---|
| SP_WC1 | The user sends a command to start or stop recording a trip |
| SP_WC2 | The mobile device sensors (GPS, accelerometer, gyroscope) transmit position and movement data to the system |
| SP_WC3 | The user manually inserts data regarding a street (e.g., street name, status) |
| SP_WC4 | The user confirms or corrects information automatically detected by the system (e.g., confirming a pothole) |
| SP_WC5 | The user inputs search parameters for a path (origin and destination) |
| SP_WC6 | The external weather service transmits current condition data (e.g., temperature, wind) to the system |

Table 1.4: Shared Phenomena – World-Controlled

**Machine-Controlled Phenomena**

| SP_MC | Description |
|---|---|
| SP_MC1 | The system visualizes available bike paths between two points on a map, highlighting those with better scores |
| SP_MC2 | The system displays trip statistics to the user (speed, distance, etc.) |
| SP_MC3 | The system presents a confirmation request to the user regarding automatically detected information |

Table 1.5: Shared Phenomena – Machine-Controlled

## 1.3. Definitions, Acronyms, Abbreviations

- BBP: Bike Buddy Project
- RASD: Requirements Analysis and Specification Document
- WP: World Phenomenon
- MP: Machine Phenomenon
- SP_WC: Shared Phenomenon - World Controlled
- SP_MC: Shared Phenomenon - Machine Controlled

## 1.4. Reference Documents

- Specification document assignment
- IEEE standard documentation for RASD
- Materials for the Software Engineering 2 course made available on WeBeep

## 1.5. Document Structure

# 2 | Overall Description

Chapter 2

# 3 Specific Requirements

## 3.1. Functional Requirements

### 3.1.1. Requirements

| ID | Description |
|---|---|
| R1 | The system shall allow new users to register an account |
| R2 | The system shall allow user not registered to access without an account if they don't want to register |
| R3 | The system shall allow registered users to log in to the application |
| R4 | The system shall allow registered users to view and update their personal profile information |
| R5 | The system shall allow users to log out of their current session |
| R6 | The system shall allow registered users to record their cycling trips |
| R7 | The system shall store recorded trips to maintain a history of the user's cycling activities |
| R8 | The system shall compute and display total distance covered for each trip |
| R9 | The system shall compute and display average speed for each trip |
| R10 | The system shall compute and display total time spent speed for each trip |
| R11 | The system shall compute and display elevation gain for each trip |
| R12 | The system shall compute and display calories spent for each trip |
| R13 | The system shall compute and display additional performance metrics for the trips |
| R14 | The system shall retrieve meteorological data (weather conditions, temperature, wind speed) from an external service and enrich the trip data when available |
| R15 | The system shall allow registered users to insert information about bike paths |
| R16 | The system shall allow registered users to make publishable information about bike paths |

*Continue in the next page...*

| ID | Description |
|---|---|
| R17 | The system shall allow registered users to add info about the status of a bike path |
| R18 | The system shall allow registered users to add the presence of relevant obstacles on the bike path |
| R19 | The system shall recognise if the users are cycling on an exists bike path |
| R20 | The system shall recognise if the users are cycling on a path where cars are rare and speed limits under 50kmh |
| R21 | The system shall calculate the distance of a path |
| R22 | The system shall allow registered users to manually insert bike path data by specifying the names of the streets included in the path |
| R23 | The system shall allow registered users to manually specify the status of the street |
| R24 | The system shall allow users to manually report the presence of obstacles on the street |
| R25 | The system shall automatically infer that the registered user is cycling based on their speed |
| R26 | The system shall collect GPS data to reconstruct the path followed by the user during the ride |
| R27 | The system shall acquire data from the mobile device's accelerometer and gyroscope to detect basic metrics |
| R28 | The system shall analyze sensor data to identify potential path irregularities |
| R29 | The system shall require the user to confirm or correct the automatically acquired data |
| R30 | The system shall allow registered users to mark their inserted data as "publishable" to make it available to the community |
| R31 | The system shall merge information about the same path from multiple users |
| R32 | The system shall prioritize data based on timeframe of execution and the number of confirmations to determine the current status |
| R33 | The system shall allow any user to search for bike paths by specifying an origin and a destination |
| R34 | The system shall find for its data bike paths where an origin and a destination are specified by any users |
| R35 | The system shall visualize the bike paths connecting the specified origin and destination on a map |
| R36 | The system shall compute a score for each path based on its status and its effectiveness in reaching the destination from the origin |

*Continue in the next page...*

| ID | Description |
|---|---|
| R37 | If multiple paths exist, from the user's request the system shall display them sorted by their calculated score |

Table 3.1: Requirements List

### 3.1.2. Use Cases Diagrams

### 3.1.3. Use Cases

In this section, they are explained and represented the main identified use cases. For each Use Case, the documentation follows a two-step structure:

- 1. Use Case Table: A structured table detailing specific fields
- 2. Diagram: A Sequence Diagram visualizing the steps described in the table

### UC1. Sign in to BBP

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.2: Sign in to BBP

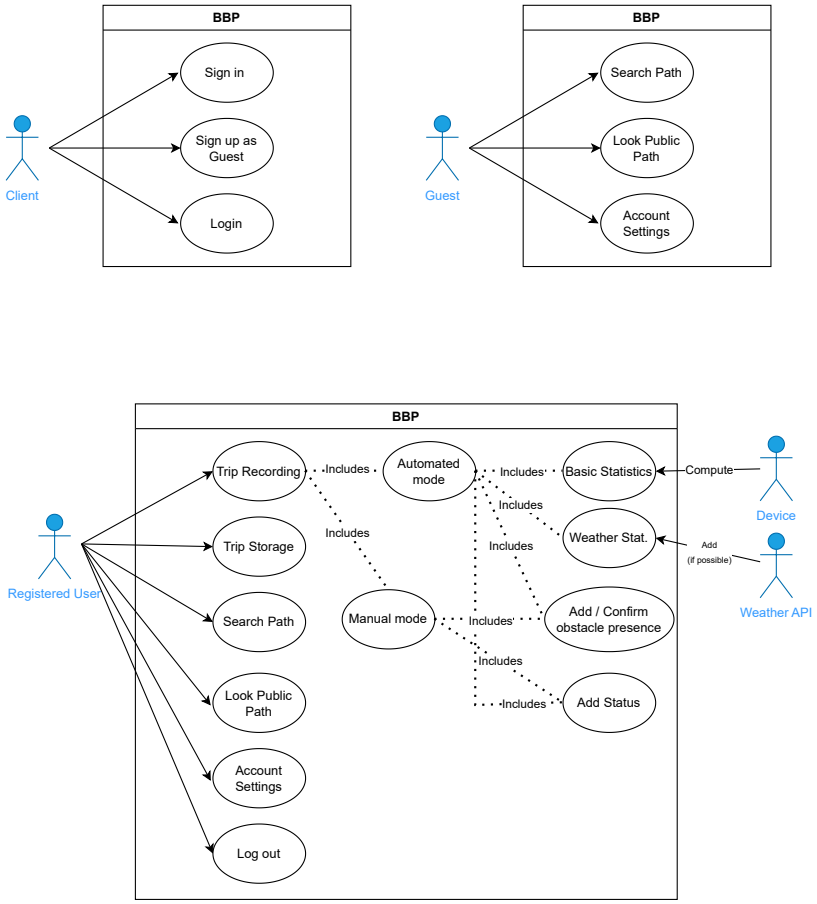



Figure 3.1: Use Cases

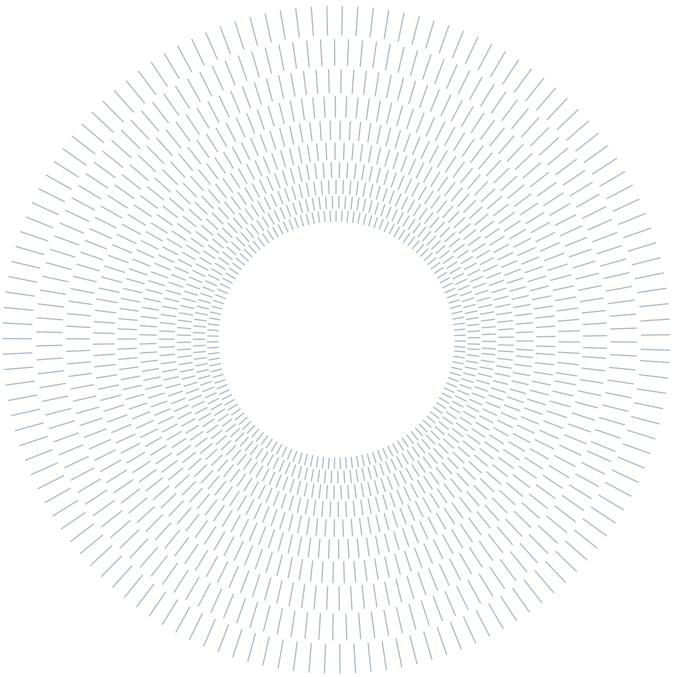


Figure 3.2: Sign in to BBP

### UC2. Login to BBP

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.3: Login to BBP

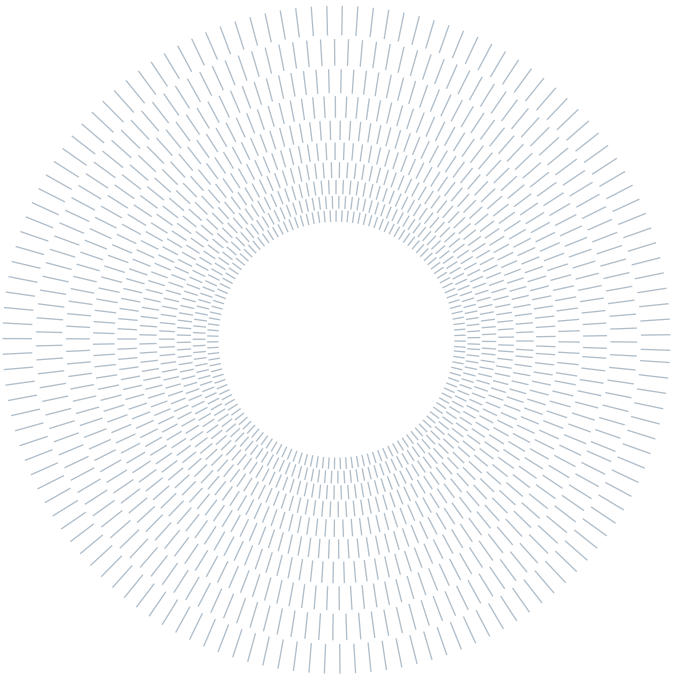


Figure 3.3: Login to BBP

### UC3. Logout to BBP

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.4: Logout to BBP

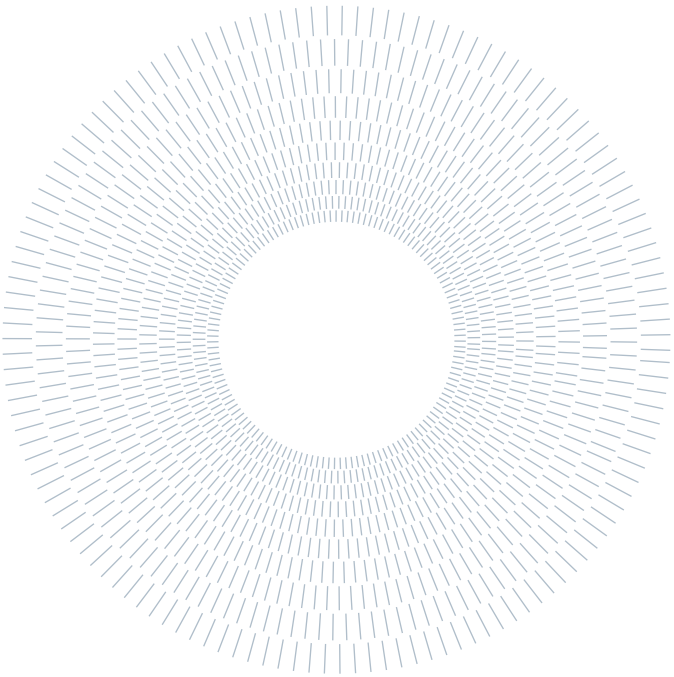


Figure 3.4: Logout to BBP

### UC4. Recording with Manual Mode

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.5: Recording with Manual Mode

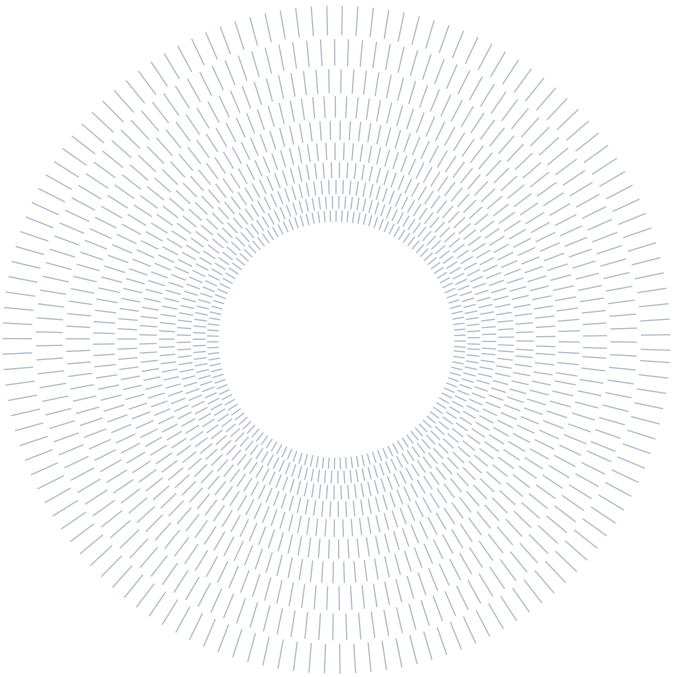


Figure 3.5: Recording with Manual Mode

### UC5. Recording with Automated Mode

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.6: Recording with Automated Mode

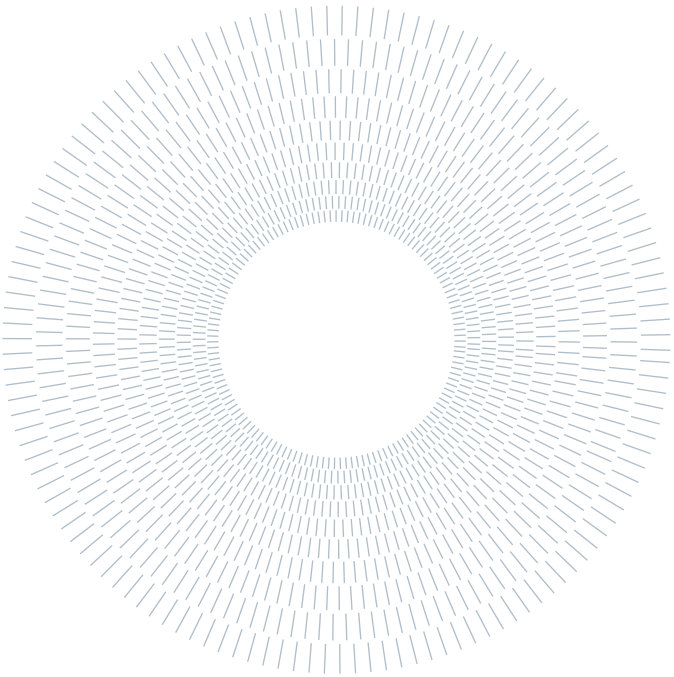


Figure 3.6: Recording with Automated Mode

### UC6. Personal Trip Storage

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.7: Personal Trip Storage

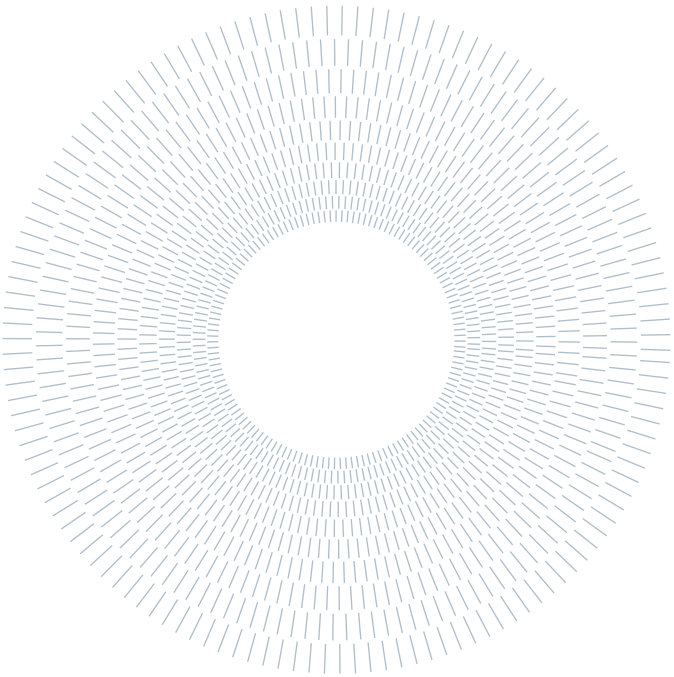


Figure 3.7: Personal Trip Storage

### UC7. Search Path

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.8: Search Path

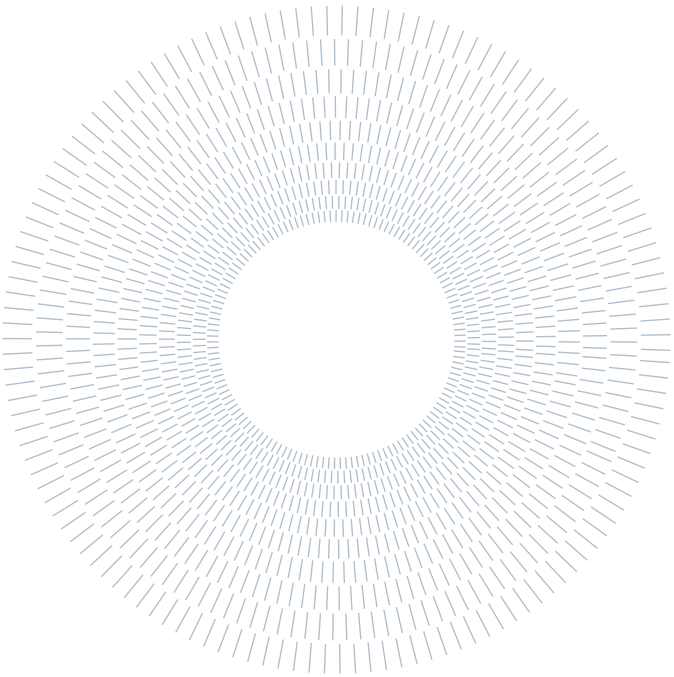


Figure 3.8: Search Path

### UC8. Look Publish Path

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.9: Look Publish Path

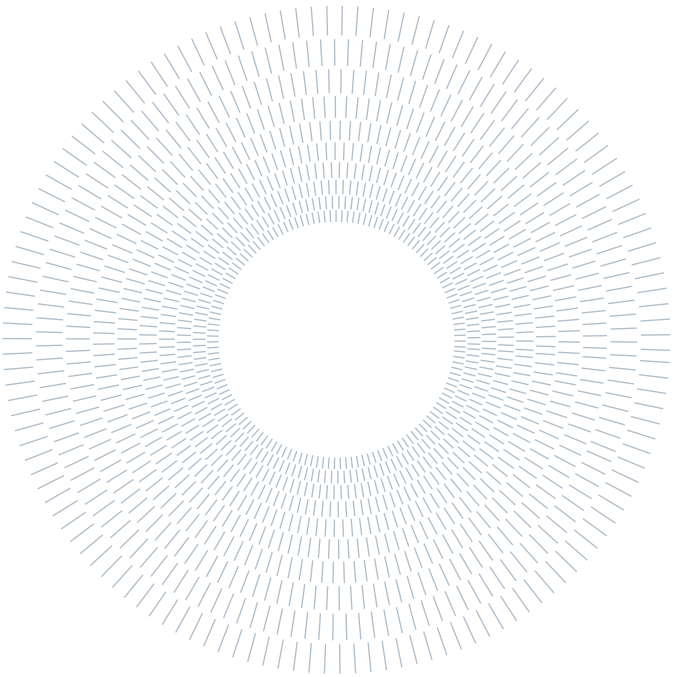


Figure 3.9: Look Publish Path

### UC9. Account Settings

| | |
|---|---|
| **Actor** | abc |
| **Entry conditions** | abc |
| **Event Flow** | • abc |
| **Exit condition** | abc |
| **Exceptions** | • abc |

Table 3.10: Account Settings

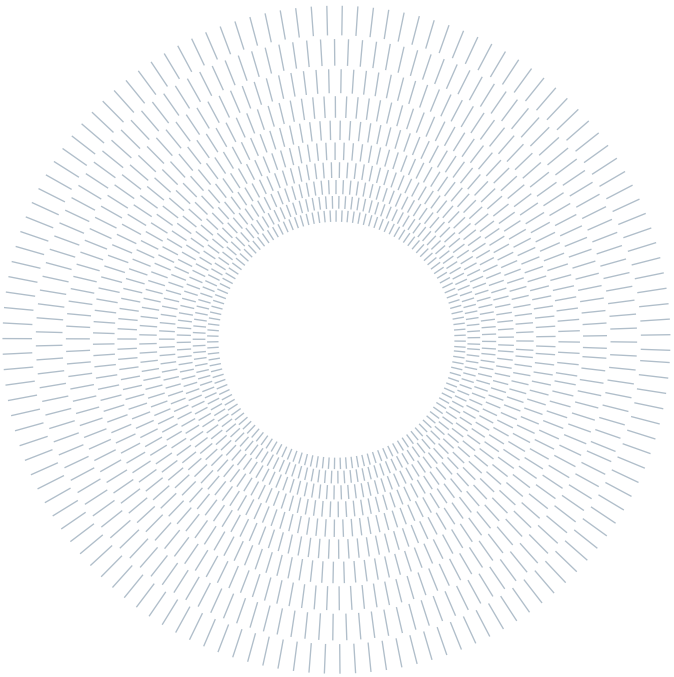


Figure 3.10: Account Settings

# 4 | Formal Analysis Using Alloy

For readability and simplicity motivations we decided to insert the most important objects of the model simplifying the complex ones (for example, individual nodes of streets are not modeled) and we kept what we thought was essential to verify simple assertions.

## 4.1. Signatures

### 4.1.1. Status

```
abstract sig Status{}
one sig Clear extends Status{}
one sig RoadWork extends Status{}
one sig Pothole extends Status{}
```

**Listing 4.1:** Status signature

An object Status is used to be associated with a street segment, therefore allowing us to better describe streets and to introduce potholes. The actual signature is abstract since we wanted to make the object expandable with more variants (we added Clear, RoadWork and Pothole as examples).

### 4.1.2. Segment

```
sig Segment{
    nextSegment: lone Segment,
    status: some Status
}{
    // a segment cannot point to itself as next
    // (but cycles are allowed)
    nextSegment != this

    // if status is clear then its the only status possible
    Clear in status implies #status = 1
}
```

**Listing 4.2:** Segment signature

An object Segment is used to model street segments: every segment is connected to at most another segment (forming what actively is a chain of streets). To have consistent segments, they are to be considered as a part of a physical street and they also model direction of said street: therefore two different segments can be the same street segment but traversed in opposite direction, this implies that a necessary condition that we need to add is that a segment cannot connect to itself.

### 4.1.3. Path

```
sig Path{
    start: Segment,
    score: Int
}{
    // Paths are saved only if associated with
    // at least a Trip
    some t: Trip| t.path = this
}
```

**Listing 4.3:** Path signature

Following the definition of a Segment, a Path is simply an objects that keeps the first segment of an arbitrarily long chain of segments. Paths also have a score and, since paths are discovered and inserted by RegisteredUsers, it is necessary that a Trip (see Trip signature) has said Path associated to it.

### 4.1.4. Statistics

```
sig Statistics{}{
    // Statistics are saved only if associated
    // with at least a trip
    some t: Trip| t.averageSpeed = this
    or t.duration = this
    or t.externalStats = this
}
```

**Listing 4.4:** Statistics signature

### 4.1.5. Trip

```
sig Trip{
    path: one Path,
    averageSpeed: one Statistics,
    duration: one Statistics,
    externalStats: lone Statistics
}{
    // statistics are different
    averageSpeed != duration and duration != externalStats

    // Trips are saved only if associated with
    // at least a registered user
    one u: RegisteredUser| this in u.registeredTrips
}
```

**Listing 4.5:** Trip signature

A Trip is an object that has one single path, the statistics associated with the trip itself inserted by the user and the statistics retrieved from the external service. Because Trips are created by the user, it is necessary that every Trip has exactly one Path and that one RegisteredUser has registered said Trip.

### 4.1.6. RegisteredUser

```
sig RegisteredUser {
    registeredTrips: set Trip,
    publishedTrips: set Trip
}{
    // published trips are a subset of all registered trips
    publishedTrips in registeredTrips
}
```

**Listing 4.6:** RegisteredUser signature

A RegisteredUser models a user with ability to record trips. We decided not to model Guests in Alloy because the functionalities we wanted to test are used only by users with an account, so the users that we used have published trips and registered trips, where the published trips are naturally a subset of the registered ones.

## 4.2. Facts

Introduction (explain why it is needed that facts are to be inserted)

### 4.2.1. Segment facts

```
fact segmentsFacts{

    // Segments are saved only if associated with another
    // segment or a Path (and at least an object Path
    // is needed in the chain)
    all s: Segment| (
        some ss: Segment|
            ss.nextSegment = s  or
            some p: Path| p.start = s)

    all s: Segment|
        // if segment is last in chain or in a cycle
        (s.nextSegment = none or s in s.^nextSegment) implies
        // then from said segment i can walk backwards
        // until i find a path's start
        (some p: Path | p.start in s.^~nextSegment)
}
```

**Listing 4.7:** Segments fact

Explain why segments need more facts

### 4.2.2. Path facts

```
fact pathsFacts{
    // Paths are equal if the segments inside
    // are the same sequence of objects
    all p1,p2: Path|
        p1 != p2 implies (
            showPath[p1] != showPath[p2])
}
```

**Listing 4.8:** Paths fact

Explain why paths need more facts

# 5 Effort Spent

A table representing the total hours of work is shown here.

| Work Areas | Alessandro Burato | Gabriele Crimella | Davide Faccini |
|---|---|---|---|
| Introduction | 1.5h | 1.5h | 0h |
| Overall Description | 0h | 0h | 0h |
| Specific Requirements | 0h | 0h | 0h |
| Alloy | 0h | 0h | 0h |
| Group Meetings and Reviews | 8h | 8h | 8h |

Table 5.1: Table of Effort

# 6 | References

Chapter 6

# List of Figures

3.1 Use Cases . . . . . 10
3.2 Sign in to BBP . . . . . 11
3.3 Login to BBP . . . . . 13
3.4 Logout to BBP . . . . . 15
3.5 Recording with Manual Mode . . . . . 17
3.6 Recording with Automated Mode . . . . . 19
3.7 Personal Trip Storage . . . . . 21
3.8 Search Path . . . . . 23
3.9 Look Publish Path . . . . . 25
3.10 Account Settings . . . . . 27

# List of Tables

1.1 List of Goals . . . . . 2
1.2 World Phenomena . . . . . 2
1.3 Machine Phenomena . . . . . 3
1.4 Shared Phenomena – World-Controlled . . . . . 3
1.5 Shared Phenomena – Machine-Controlled . . . . . 4

3.1 Requirements List . . . . . 9
3.2 Sign in to BBP . . . . . 9
3.3 Login to BBP . . . . . 12
3.4 Logout to BBP . . . . . 14
3.5 Recording with Manual Mode . . . . . 16
3.6 Recording with Automated Mode . . . . . 18
3.7 Personal Trip Storage . . . . . 20
3.8 Search Path . . . . . 22
3.9 Look Publish Path . . . . . 24
3.10 Account Settings . . . . . 26

5.1 Table of Effort . . . . . 33